MILITARY SPECIFICATION

WHEEL SET, ABRASIVE, RUBBER POINTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers abrasive-impregnated rubber points and detachable mandrels for use with straight dental handpieces.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	-	Box, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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MIL-P-116 - Methods of Preservation.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-vapor-proof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED TEST METHOD STD-601 - Rubber: Sampling and testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B74.10 Specification for Grading of Abrasive Microgrits.

ANSI B74.12 Specification for the Size of Abrasive Grain - Grinding Wheels, Polishing, and General Industrial uses.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 Materials. Materials shall conform in all respects to applicable specifications and as hereinafter specified. All other materials not covered by specifications shall be of high quality throughout, and suitable in all respects for the purpose intended.

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3.1.1 Abrasive. The abrasive shall be silicon carbide consisting of 4 parts Grit No. 320 and 6 parts Grit No. 240 by weight. Grit numbers shall be controlled within the limits for sizing as specified in ANSI Standards B74.10 and B74.12, respectively, when tested as specified in 4.4.1.

3.1.2 Bonding agent. The bonding material shall be age-resistant synthetic rubber.

3.1.3 Steel. Mandrels shall be made of corrosion-resisting steel, carbon steel, or alloy steel, suitable for the purpose intended.

3.2 Design and construction. Shall be a set of silicon carbide abrasive-impregnated rubber points and detachable mandrels for use with standard straight handpieces. The rubber points shall be suitable for polishing and grinding of chromium and stainless steel dental alloys.

3.2.1 Points. Points shall be fabricated of rubber, impregnated with silicon carbide abrasive as specified in 3.1.1. Points shall conform to figure 1. They shall be cylindrical in shape with square edges. A hole, extending from the base end toward the center and running parallel to the long axis, shall accommodate the mandrels (3.2.2) furnished with the points. The hole shall be centered within .002 inch of the centerline of the point.

3.2.1.1 Accelerated aging. Points shall not become hard, brittle, soft, nor tacky when tested as specified in 4.4.2.

3.2.1.2 Hardness. Points shall be within the Shore Durometer, Scale A, range of hardness of 90 to 100 when tested as specified in 4.4.3.

3.2.2 Mandrels. Mandrels shall be fabricated of steel (see 3.1.3). Mandrels shall conform to either of the designs illustrated in figure 1. The operative portion shall be threaded (44 to 52 threads per inch) for insertion into rubber points and shall engage and hold points firmly and concentrically when in use. The thread may have a longitudinal groove, or "flute". The operative end, excluding the threaded portion, shall be pointed. The non-operative end of the mandrel shall be designed for use in standard, straight dental engine handpieces. The mandrel shall have a shoulder, or a fixed or movable nut, to prevent the mandrel from penetrating too deeply into the point.

3.2.2.1 Finish. Mandrels shall be smoothly finished. Carbon and alloy steel mandrels shall be nickel or chromium plated.

3.2.2.2 Corrosion resistance. Mandrels shall show no more than 2 percent corrosion when tested as specified in 4.4.4.

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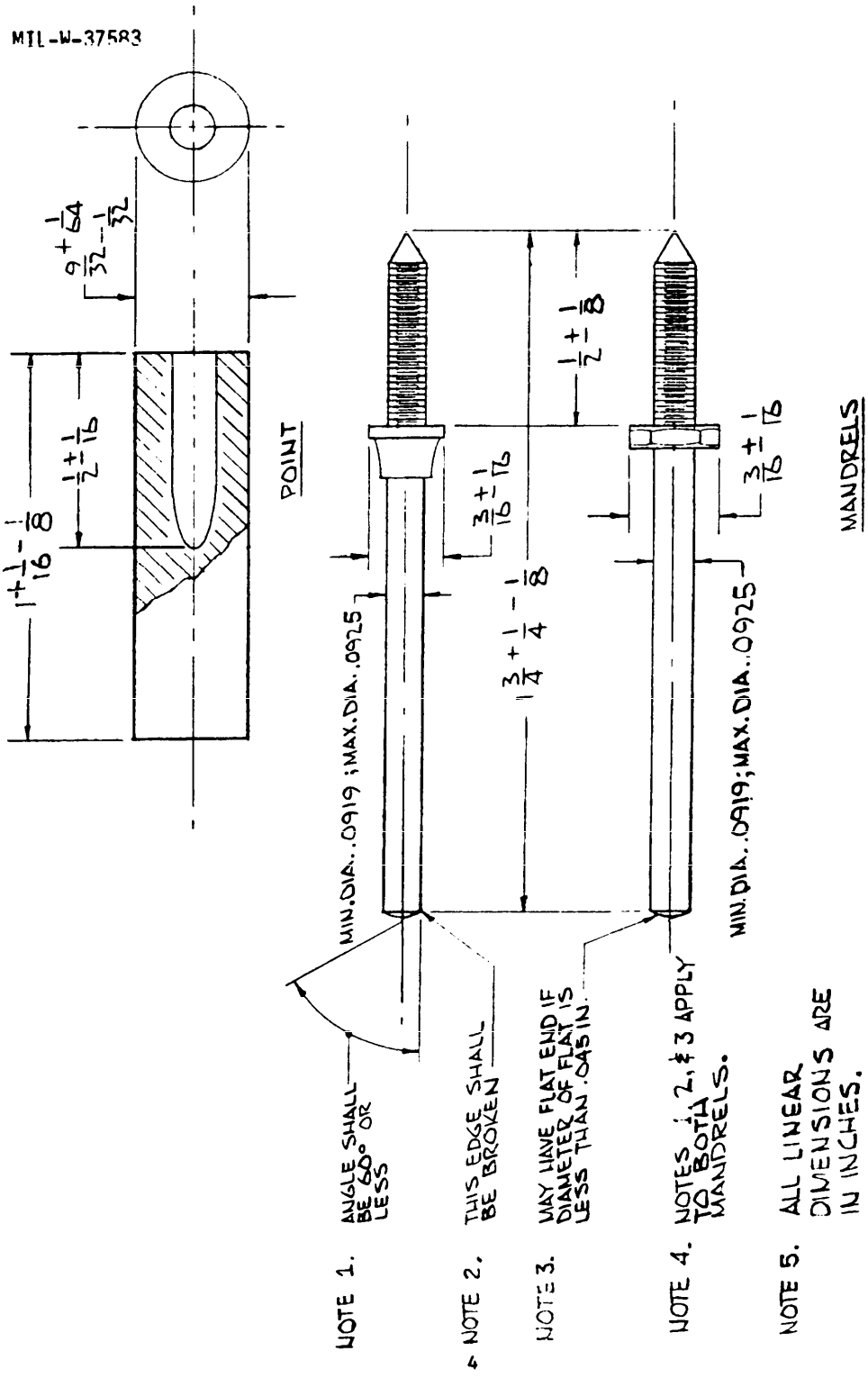


FIGURE 1. Wheel Set, Abrasive.

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3.2.2.3 Runout. The total indicated runout for the mandrels shall not exceed .005 inch when tested as specified in 4.4.5.

3.2.3 Count. The average number of points per set shall be not less than 100, with no less than 97 in any individual set. Each set shall contain a minimum of 2 mandrels.

3.3 Identification marking. Identification marking of mandrel with the name or registered trade mark of the manufacturer is optional.

3.4 Workmanship. The points shall be symmetrical, smoothly and properly finished, and shall be free from defects which detract from their appearance or may impair their serviceability. Points shall be free of dirt, grease, oil, or foreign matter. Mandrels shall have no burrs or unspecified sharpness. Mandrel surface shall not be stained or discolored. Plated mandrel surfaces shall exhibit no evidence of peeling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Inspection shall include the examination of 4.3 and tests of 4.4.

4.1.3 Certificates of compliance. Certificates of compliance for the identity of the abrasive grit sizes, including the ratio of each abrasive grit size, may be furnished in lieu of actual performance of such testing by the contractor. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Sampling. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table I. The unit of product for sampling purposes shall be one point or one mandrel, as indicated.

TABLE I. Sampling for examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional and count examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. Sampling for tests.

Component (Unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Raw abrasive	Grit size	3.1.1	4.4.1	1/	2/
Point	Accelerated aging	3.2.1.1	4.4.2	S-1	1.0
	Hardness	3.2.1.2	4.4.3	S-1	1.5
Mandrel	Corrosion resistance	3.2.2.2	4.4.4	S-1	1.0
	Runout	3.2.2.3	4.4.5	S-1	1.5

1/ Raw materials. A sufficient quantity of each lot of each abrasive used in the manufacture of the points shall be selected for grit size analysis.

2/ Acceptance number shall be zero.

4.3 Examination. The wheel sets shall be examined to determine compliance with all requirements contained in this specification.

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4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table III. Examination shall not be restricted to the classified possible defects listed below:

TABLE III. Classification of defects.

Major:	
	Point
101	Shape not as specified.
102	Surface ruptured.
103	Point not complete, i.e., material missing.
104	Unspecified abrasive matter imbedded.
	Mandrel
111	Shape not as specified.
112	Does not fit points properly.
113	Burrs, or unspecified sharpness.
114	Presence of rust.
115	Plating peeling (for plated mandrels)
116	Threads not correctly fashioned.
Minor:	
	Point
201	Surface not of uniform color.
202	Surface not clean, i.e., foreign matter present.
	Mandrel
211	Finish not as specified.
212	Surface stained or discolored.

4.3.2 Dimensional and count examination. The points and mandrels shall be examined for defects in dimensions. Any dimension, including component count (see 3.2.3), not within the tolerances specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129."

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Abrasive size. The size of the silicon-carbide abrasive grit shall be determined in accordance with ANSI Standard B74.10 for Grit No. 320 and ANSI Standard B74.12 for Grit No. 240.

4.4.2 Accelerated aging. Points shall be subjected to the aging test specified in Method 7111 of Fed.-Test-Method-Std. 601.

4.4.3 Hardness. Points shall be subjected to the durometer hardness test specified in Method 3021 of Fed-Test-Method-Std. 601.

4.4.4 Boil test. Mandrels shall be scrubbed with soap and water, rinsed thoroughly in distilled water, and dried. After cleaning, mandrels shall be boiled in distilled water in a glass beaker for 30 minutes and allowed to remain submerged for an additional 24 hours.

4.4.5 Runout. The mandrel shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to .0002 inch and equipped with a knife-edge foot, or a shadow-graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the mandrel, and so positioned that the foot contacts the mandrel within 1/8 inch of the point. The maximum and minimum readings obtained during one complete 360-degree rotation of the mandrel shall be measured and recorded to the nearest .0005 inch. The difference between these two readings shall be taken as the total indicated runout.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One set consisting of 100 abrasive rubber points and two detachable mandrels as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit packaging. Each set shall be packaged in accordance with MIL-P-116, Method III, utilizing a box as the container.

5.1.1.2 Level C. Each set shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

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5.2.1 Level A. Forty eight (48) units of issue shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Forty eight (48) units of issue shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit and exterior container. Each unit and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The wheel sets are intended for use in dental laboratories.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-299-8204	WHEEL SET, ABRASIVE, RUBBER, STRAIGHT HANDPIECE, SET OF 100.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DOD - MB

Preparing activity:
DLA-DM

Project No. 6520-1434

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-W-37583 WHEEL SET, ABRASIVE, RUBBER POINTS

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

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